

UNIT 3: CSS – Cascading Style Sheet

3.1 Introduction

CSS is an abbreviation for **Cascading Style Sheets**. CSS works with HTML and other Markup Languages (such as XHTML and XML) to control the way the content is presented.

It is a language that describes the style of an HTML document. This concerns features such as typeface, **background, link colors, margins, and placement of objects** on a page. Display instructions for these features are called “**style rules**” in CSS lingo. Cascading Style Sheets are a means to separate the appearance of a webpage from the content of a webpage. CSS is a recommendation of the World Wide Web Consortium (the W3C). The W3C is a consortium of web stakeholders: universities, companies such as Microsoft, Netscape and Macromedia, and experts in many web related fields. The presentation is specified by styles, which are presented in a style sheet.

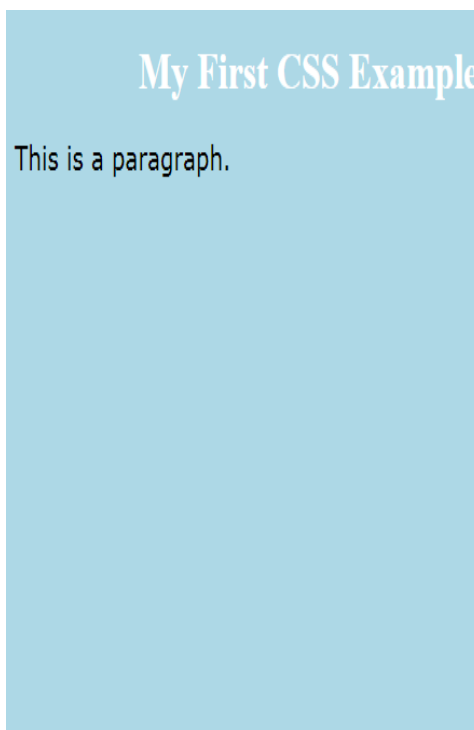
CSS Example

```
<!DOCTYPE html>
<html>
<head>
<style>
body {
    background-color: lightblue;
}

h1 {
    color: white;
    text-align: center;
}

p {
    font-family: verdana;
    font-size: 20px;
}
</style>
</head>
<body>
<h1>My First CSS Example</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

Output



3.2 Why Use CSS?

CSS is used to define styles for your web pages, including the design, layout and variations in display for different devices and screen sizes.

- CSS Solved a Big Problem

- ✓ HTML was NEVER intended to contain tags for formatting a web page.
- ✓ HTML was created to **describe the content** of a web page, like:

```
<h1>This is a heading. </h1>
```

```
<p>This is a paragraph. </p>
```

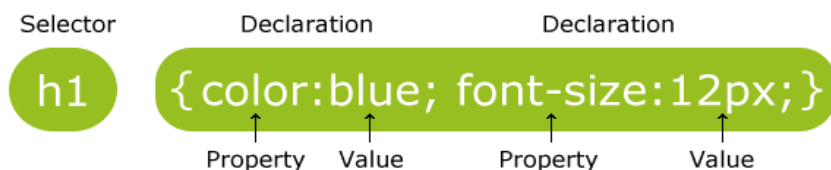
When tags like `<font>`, and color attributes were added to the HTML, it started a nightmare for web developers. Development of large websites, where fonts and color information were added to every single page, became a long and expensive process. To solve this problem, the World Wide Web Consortium (W3C) created CSS. CSS removed the style formatting from the HTML page.

- CSS Saves a Lot of Work.

The style definitions are normally saved in external .css files. With an external style sheet file, you can change the look of an entire website by changing just one file!

3.3 CSS Syntax

A CSS ruleset consists of a selector and a declaration block:



- The selector points to the HTML element you want to style.
- The declaration block contains one or more declarations separated by semicolons.
- Each declaration includes a CSS property name and a value, separated by a colon.

A CSS declaration always ends with a semicolon, and declaration blocks are surrounded by curly braces.

- CSS Selectors

Three types of CSS Selectors

- **Element**
- **Class**
- **ID**

In the following example all <p> elements will be center-aligned, with a red text color:

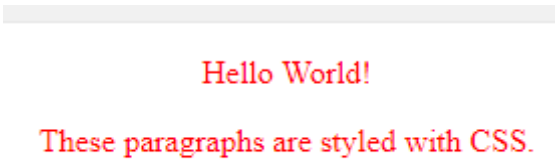
Example

```
p {
    color: red;
    text-align: center;
}
```

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
    color: red;
    text-align: center;
}
</style>
</head>
<body>

<p>Hello World!</p>
<p>These paragraphs are styled with CSS.</p>

</body>
</html>
```



CSS Selectors

CSS selectors are used to "find" (or select) HTML elements based on their element name, id, class, attribute, and more.

The element Selector

The element selector selects elements based on the element name. You can select all <p> elements on a page like this (in this case, all <p> elements will be center-aligned, with a red text color):

Example

```
p {
  text-align: center;
  color: red;
}
```

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  text-align: center;
  color: red;
}
</style>
</head>
<body>

<p>Every paragraph will be affected by the style.</p>
<p id="para1">Me too!</p>
<p>And me!</p>

</body>
</html>
```

Every paragraph will be affected by the style.

Me too!

And me!

The id Selector

The id selector uses the id attribute of an HTML element to select a specific element. The id of an element should be unique within a page, so the id selector is used to select one unique element. To select an element with a specific id, write a hash (#) character, followed by the id of the element.

The style rule below will be applied to the HTML element with id="para1":

Example

```
#para1 {
  text-align: center;
  color: red;
}
```

```

<!DOCTYPE html>
<html>
<head>
<style>
#para1 {
    text-align: center;
    color: red;
}
</style>
</head>
<body>
Hello World!

<p id="para1">Hello World!</p>
<p>This paragraph is not affected by the style.</p>    This paragraph is not affected by the style.

</body>
</html>

```

Notes: An id name cannot start with a number

The class Selector

The class selector selects elements with a specific class attribute. To select elements with a specific class, write a period (.) character, followed by the name of the class. In the example below, all HTML elements with class="center" will be red and center-aligned:

Example

```

.center {
    text-align: center;
    color: red;
}

```

```
<!DOCTYPE html>
<html>
<head>
<style>
.center {
  text-align: center;
  color: red;
}
</style>
</head>
<body>
```

Red and center-aligned heading

```
<h1 class="center">Red and center-aligned heading</h1>
<p class="center">Red and center-aligned paragraph.</p>

</body>
</html>
```

Red and center-aligned paragraph.

You can also specify that only specific HTML elements should be affected by a class. In the example below, only `<p>` elements with `class="center"` will be center-aligned:

Example

```
p.center {
  text-align: center;
  color: red;
}
```

```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: red;
}
</style>
</head>
<body>
```

```
<h1 class="center">This heading will not be affected</h1>
<p class="center">This paragraph will be red and center-aligned.
</p>

</body>
</html>
```

This heading will not be affected

This paragraph will be red and center-aligned.

HTML elements can also refer to more than one class. In the example below, the <p> element will be styled according to class="center" and to class="large":

Example

<p class="center large">This paragraph refers to two classes.</p>

```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: red;
}
```

```
p.large {
  font-size: 300%;
}
</style>
</head>
<body>
```

This heading will not be affected

This paragraph will be red and center-aligned.

```
<h1 class="center">This heading will not be affected</h1>
<p class="center">This paragraph will be red and center-aligned
</p>
<p class="center large">This paragraph will be red, center-
aligned, and in a large font-size.</p>

</body>
</html>
```

This paragraph will be red,
center-aligned, and in a
large font-size.

Note: A class name cannot start with a number.

- Grouping Selectors

If you have elements with the same style definitions, like this:

```

h1 {
    text-align: center;
    color: red;
}
h2 {
    text-align: center;
    color: red;
}
p {
    text-align: center;
    color: red;
}

```

It will be better to group the selectors, to minimize the code. To group selectors, separate each selector with a comma. In the example below we have grouped the selectors from the code above:

Example

```

h1, h2, p {
    text-align: center;
    color: red;
}

```

```

<!DOCTYPE html>
<html>
<head>
<style>
h1, h2, p {
    text-align: center;
    color: red;
}
</style>
</head>
<body>

<h1>Hello World!</h1>
<h2>Smaller heading!</h2>
<p>This is a paragraph.</p>

</body>
</html>

```

Hello World!

Smaller heading!

This is a paragraph.

3.4 CSS Comments

Comments are used to explain the code, and may help when you edit the source code at a later date. Comments are ignored by browsers. A CSS comment starts with `/*` and ends with `*/`. Comments can also span multiple lines:

Example

```
p {
    color: red;
    /* This is a single-line comment */
    text-align: center;
}
/* This is
a multi-line
comment */
```

3.5 Three Ways to Insert CSS

There are three ways of inserting a style sheet:

- External style sheet
- Internal style sheet
- Inline style

- **External Style Sheet**

With an external style sheet, you can change the look of an entire website by changing just one file. Each page must include a reference to the external style sheet file inside the <link> element. The <link> element goes inside the <head> section:

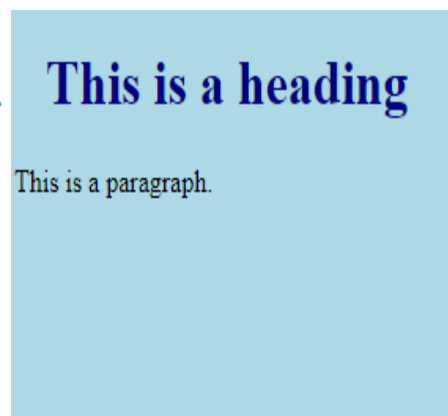
Example

```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```



An external style sheet can be written in any text editor. The file should not contain any html tags. The style sheet file must be saved with a .css extension. Here is how the "mystyle.css" looks:

```
body {
    background-color: lightblue;
}
h1 {
    color: navy;
    margin-left: 20px;
}
```

- Internal Style Sheet

An internal style sheet may be used if one single page has a unique style. Internal styles are defined within the <style> element, inside the <head> section of an HTML page:

Example

```
<head>
<style>
body {
    background-color: linen;
}
h1 {
    color: maroon;
    margin-left: 40px;
}
</style>
</head>
```

- Inline Styles

An inline style may be used to apply a unique style for a single element. To use inline styles, add the style attribute to the relevant element. The style attribute can contain any CSS property. The example below shows how to change the color and the left margin of a <h1> element:

Example

```
<h1 style="color:blue;margin-left:30px;">This is a heading</h1>
```

```

<!DOCTYPE html>
<html>
<body>

<h1 style="color:blue;margin-left:30px;">This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>

```

This is a heading

This is a paragraph.

3.6 Multiple Style Sheets

If some properties have been defined for the same selector (element) in different style sheets, the value from the last read style sheet will be used.

Example

Assume that an external style sheet has the following style for the <h1> element:

```

h1 {
    color: navy;
}

```

then, assume that an internal style sheet also has the following style for the <h1> element:

```

h1 {
    color: orange;
}

```

If the internal style is defined after the link to the external style sheet, the <h1> elements will be "orange":

Example

```

<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
<style>
h1 {
    color: orange;
}
</style>
</head>

```

However, if the internal style is defined before the link to the external style sheet, the <h1> elements will be "navy":

Example

```

<head>
<style>
h1 {
    color: orange;
}
</style>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>

```

3.7 Cascading Order

What style will be used when there is more than one style specified for an HTML element? Generally speaking we can say that all the styles will "cascade" into a new "virtual" style sheet by the following rules, where number one has the highest priority:

1. Inline style (inside an HTML element)
2. External and internal style sheets (in the head section)
3. Browser default

So, an inline style (inside a specific HTML element) has the highest priority, which means that it will override a style defined inside the <head> tag, or in an external style sheet, or a browser default value.

3.8 CSS Colors

HTML colors can be specified using predefined color names, RGB, HEX, HSL, RGBA, and HSLA values.

- **Background Color**

You can set the background color for HTML elements:

Example

```

<h1 style="background-color:DodgerBlue">Hello World</h1>
<p style="background-color:Tomato">Lorem ipsum...</p>

```

- **Text Color**

You can set the color of text:

```
<h1 style="color:Tomato">Hello World</h1>
<p style="color:DodgerBlue">Lorem ipsum...</p>
<p style="color:MediumSeaGreen">Ut wisi ad...</p>
```

- **Border Color**

You can set the color of borders:

Hello World

Hello World

Hello World

Example

```
<h1 style="border:2px solid Tomato">Hello World</h1>
<h1 style="border:2px solid DodgerBlue">Hello World</h1>
<h1 style="border:2px solid Violet">Hello World</h1>
```

- **RGB Value**

In HTML, a color can be specified as an RGB value, using this formula:

rgb(*red*, *green*, *blue*)

Each parameter (red, green, and blue) defines the intensity of the color between 0 and 255. For example, rgb(255, 0, 0) is displayed as red, because red is set to its highest value (255) and the others are set to 0.

- **HEX Value**

In HTML, a color can be specified using a hexadecimal value in the form:

#rrggbb

Where rr (red), gg (green) and bb (blue) are hexadecimal values between 00 and ff (same as decimal 0-255). For example, #ff0000 is displayed as red, because red is set to its highest value (ff) and the others are set to the lowest value (00).

- **HSL Value**

In HTML, a color can be specified using hue, saturation, and lightness (HSL) in the form:

hsl(*hue*, *saturation*, *lightness*)

Hue is a degree on the color wheel from 0 to 360. 0 is red, 120 is green, and 240 is blue. Saturation is a percentage value, 0% means a shade of gray, and 100% is the full color. Lightness is also a percentage, 0% is black, 50% is neither light nor dark, 100% is white

- **RGBA Value**

RGBA color values are an extension of RGB color values with an alpha channel - which specifies the opacity for a color. An RGBA color value is specified with:

`rgba(red, green, blue, alpha)`

The alpha parameter is a number between 0.0 (fully transparent) and 1.0 (not transparent at all):

- **HSLA Value**

HSLA color values are an extension of HSL color values with an alpha channel - which specifies the opacity for a color. An HSLA color value is specified with:

`hsla(hue, saturation, lightness, alpha)`

The alpha parameter is a number between 0.0 (fully transparent) and 1.0 (not transparent at all):

CSS Backgrounds

The CSS background properties are used to define the background effects for elements.

CSS background properties:

- Background-color
- Background-image
- Background-repeat
- Background-attachment
- Background-position

- **Background Color**

The **background-color** property specifies the background color of an element. The background color of a page is set like this:

Example

```
body {
    background-color: lightblue;
}
```

With CSS, a color is most often specified by:

- a valid color name - like "red"
- a HEX value - like "#ff0000"
- an RGB value - like "rgb(255,0,0)"

In the example below, the <h1>, <p>, and <div> elements have different background colors:

Example

```
h1 {
    background-color: green;
}
div {
    background-color: lightblue;
}
p {
    background-color: yellow;
}
```

3.9 Background Image

The **background-image** property specifies an image to use as the background of an element. By default, the image is repeated so it covers the entire element. The background image for a page can be set like this:

Example

```
body {
    background-image: url("paper.gif");
}
```

Below is an example of a **bad** combination of text and background image. The text is hardly readable:

Example

```
body {
    background-image: url("bgdesert.jpg");
}
```

Note: When using a background image, use an image that does not disturb the text.

Background Image - Repeat Horizontally or Vertically

By default, the `background-image` property repeats an image both horizontally and vertically. Some images should be repeated only horizontally or vertically, or they will look strange, like this:

Example

```
body {
    background-image: url("gradient_bg.png");
}
```

If the image above is repeated only horizontally (`background-repeat: repeat-x;`), the background will look better:

Example

```
body {
    background-image: url("gradient_bg.png");
    background-repeat: repeat-x;
}
```

Tip: To repeat an image vertically, set `background-repeat: repeat-y;`

Background Image - Set position and no-repeat

Showing the background image only once is also specified by the `background-repeat` property:

Example

```
body {
    background-image: url("img_tree.png");
    background-repeat: no-repeat;
}
```

In the example above, the background image is shown in the same place as the text. We want to change the position of the image, so that it does not disturb the text too much. The position of the image is specified by the `background-position` property:

Example

```
body {
    background-image: url("img_tree.png");
    background-repeat: no-repeat;
}
```

```
    background-position: right top;
}
```

Background Image - Fixed position

To specify that the background image should be fixed (will not scroll with the rest of the page), use the `background-attachment` property:

Example

```
body {
    background-image: url("img_tree.png");
    background-repeat: no-repeat;
    background-position: right top;
    background-attachment: fixed;
}
```

Background - Shorthand property

To shorten the code, it is also possible to specify all the background properties in one single property. This is called a shorthand property. The shorthand property for background is `background`:

Example

```
body {
    background: #ffffff url("img_tree.png") no-repeat right top;
}
```

When using the shorthand property the order of the property values is:

- `background-color`
- `background-image`
- `background-repeat`
- `background-attachment`
- `background-position`

It does not matter if one of the property values is missing, as long as the other ones are in this order.

3.10 CSS Margins

The CSS `margin` properties are used to generate space around elements. The `margin` properties set the size of the white space outside the border. With CSS, you have full control over the

margins. There are CSS properties for setting the margin for each side of an element (top, right, bottom, and left).

- Margin - Individual Sides

CSS has properties for specifying the margin for each side of an element:

- `margin-top`
- `margin-right`
- `margin-bottom`
- `margin-left`

All the margin properties can have the following values:

- `auto` - the browser calculates the margin
- `length` - specifies a margin in px, pt, cm, etc.
- `%` - specifies a margin in % of the width of the containing element
- `inherit` - specifies that the margin should be inherited from the parent element

The following example sets different margins for all four sides of a `<p>` element:

Example

```
p {
  margin-top: 100px;
  margin-bottom: 100px;
  margin-right: 150px;
  margin-left: 80px;
}
```

- Margin - Shorthand Property

To shorten the code, it is possible to specify all the margin properties in one property. The `margin` property is a shorthand property for the following individual margin properties:

- `margin-top`
- `margin-right`
- `margin-bottom`
- `margin-left`

Example

```
p {
  margin: 100px 150px 100px 80px;
}
```

If the `margin` property has four values:

- **`margin: 25px 50px 75px 100px;`**
 - top margin is 25px
 - right margin is 50px
 - bottom margin is 75px
 - left margin is 100px

If the `margin` property has three values:

- **`margin: 25px 50px 75px;`**
 - top margin is 25px
 - right and left margins are 50px
 - bottom margin is 75px

If the `margin` property has two values:

- **`margin: 25px 50px;`**
 - top and bottom margins are 25px
 - right and left margins are 50px

If the `margin` property has one value:

- **`margin: 25px;`**
 - all four margins are 25px

- The auto Value

You can set the margin property to `auto` to horizontally center the element within its container. The element will then take up the specified width, and the remaining space will be split equally between the left and right margins:

Example

```
div {  
  width: 300px;  
  margin: auto;  
  border: 1px solid red;  
}
```

- The inherit Value

This example lets the left margin be inherited from the parent element:

Example

```
div.container {
  border: 1px solid red;
  margin-left: 100px;
}
p.one {
  margin-left: inherit;
}
```

- Margin Collapse

Top and bottom margins of elements are sometimes collapsed into a single margin that is equal to the largest of the two margins. This does not happen on left and right margins. Only top and bottom margins. Look at the following example:

```
h1 {
  margin: 0 0 50px 0;
}

h2 {
  margin: 20px 0 0 0;
}
```

In the example above, the <h1> element has a bottom margin of 50px. The <h2> element has a top margin set to 20px. Common sense would seem to suggest that the vertical margin between the <h1> and the <h2> would be a total of 70px (50px + 20px). But due to margin collapse, the actual margin ends up being 50px.

3.11 All CSS Margin Properties

Property	Description
margin	A shorthand property for setting the margin properties in one declaration
margin-bottom	Sets the bottom margin of an element
margin-left	Sets the left margin of an element

margin-right	Sets the right margin of an element
margin-top	Sets the top margin of an element

3.12 CSS Height and Width

Setting height and width

The `height` and `width` properties are used to set the height and width of an element. The `height` and `width` can be set to `auto` (this is default. Means that the browser calculates the height and width), or be specified in *length values*, like `px`, `cm`, etc., or in percent (%) of the containing block.

This element has a height of 200 pixels and a width of 50%

Example

```
div {
  height: 200px;
  width: 50%;
  background-color: powderblue;
}
```

This element has a height of 100 pixels and a width of 500 pixels.

Example

```
div {
  height: 100px;
  width: 500px;
  background-color: powderblue;
}
```

Setting max-width

The `max-width` property is used to set the maximum width of an element. The `max-width` can be specified in *length values*, like `px`, `cm`, etc., or in percent (%) of the containing block, or set to `none` (this is default. Means that there is no maximum width). The problem with the `<div>` above occurs when the browser window is smaller than the width of the element (500px). The browser then adds a horizontal scrollbar to the page. Using `max-width` instead, in this situation, will

improve the browser's handling of small windows. The following example shows a `<div>` element with a height of 100 pixels and a max-width of 500 pixels:

```
div {
  max-width: 500px;
  height: 100px;
  background-color: powderblue;
}
```

3.13 CSS Text

Text Color

The `color` property is used to set the color of the text. With CSS, a color is most often specified by:

- a color name - like "red"
- a HEX value - like "#ff0000"
- an RGB value - like "rgb(255,0,0)"

The default text color for a page is defined in the body selector.

Example

```
body {
  color: blue;
}
h1 {
  color: green;
}
```

Text Alignment

The `text-align` property is used to set the horizontal alignment of a text. A text can be left or right aligned, centered, or justified. The following example shows center aligned, and left and right aligned text (left alignment is default if text direction is left-to-right, and right alignment is default if text direction is right-to-left):

Example

```
h1 {
  text-align: center;
}
h2 {
  text-align: left;
}
```

```
h3 {  
    text-align: right;  
}
```

When the `text-align` property is set to "justify", each line is stretched so that every line has equal width, and the left and right margins are straight (like in magazines and newspapers):

Example

```
div {  
    text-align: justify;  
}
```

3.14 Text Decoration

The `text-decoration` property is used to set or remove decorations from text. The value `text-decoration: none;` is often used to remove underlines from links:

Example

```
a {  
    text-decoration: none;  
}
```

The other `text-decoration` values are used to decorate text:

Example

```
h1 {  
    text-decoration: overline;  
}  
h2 {  
    text-decoration: line-through;  
}  
h3 {  
    text-decoration: underline;  
}
```

Note: It is not recommended to underline text that is not a link, as this often confuses the reader.

3.15 Text Transformation

The `text-transform` property is used to specify uppercase and lowercase letters in a text. It can be used to turn everything into uppercase or lowercase letters, or capitalize the first letter of each word:

Example

```
p.uppercase {  
    text-transform: uppercase;  
}  
p.lowercase {  
    text-transform: lowercase;  
}  
p.capitalize {  
    text-transform: capitalize;  
}
```

Text Indentation

The `text-indent` property is used to specify the indentation of the first line of a text:

Example

```
p {  
    text-indent: 50px;  
}
```

3.16 Letter Spacing

The `letter-spacing` property is used to specify the space between the characters in a text. The following example demonstrates how to increase or decrease the space between characters:

Example

```
h1 {  
    letter-spacing: 3px;  
}  
h2 {  
    letter-spacing: -3px;  
}
```

Line Height

The `line-height` property is used to specify the space between lines:

Example

```
p.small {  
    line-height: 0.8;  
}  
  
p.big {  
    line-height: 1.8;  
}
```

Text Direction

The `direction` property is used to change the text direction of an element:

Example

```
p {  
    direction: rtl;  
}
```

Word Spacing

The `word-spacing` property is used to specify the space between the words in a text. The following example demonstrates how to increase or decrease the space between words:

Example

```
h1 {  
    word-spacing: 10px;  
}  
h2 {  
    word-spacing: -5px;  
}
```

Text Shadow

The `text-shadow` property adds shadow to text. The following example specifies the position of the horizontal shadow (3px), the position of the vertical shadow (2px) and the color of the shadow (red):

Example

```
h1 {
  text-shadow: 3px 2px red;
}
```

CSS Fonts

The CSS font properties define the font family, boldness, size, and the style of a text.

Difference between Serif and Sans-serif Fonts



CSS Font Families

In CSS, there are two types of font family names:

- **generic family** - a group of font families with a similar look (like "Serif" or "Monospace")
- **font family** - a specific font family (like "Times New Roman" or "Arial")

3.17 Font Family

The font family of a text is set with the `font-family` property. The `font-family` property should hold several font names as a "fallback" system. If the browser does not support the first font, it tries the next font, and so on. Start with the font you want, and end with a generic family, to let the browser pick a similar font in the generic family, if no other fonts are available.

Note: If the name of a font family is more than one word, it must be in quotation marks, like: "Times New Roman".

More than one font family is specified in a comma-separated list:

Example

```
p {
  font-family: "Times New Roman", Times, serif;
}
```

Font Style

The `font-style` property is mostly used to specify italic text. This property has three values:

- normal - The text is shown normally
- italic - The text is shown in italics
- oblique - The text is "leaning" (oblique is very similar to italic, but less supported)

Example

```
p.normal {
  font-style: normal;
}
p.italic {
  font-style: italic;
}
p.oblique {
  font-style: oblique;
}
```

Font Size

The `font-size` property sets the size of the text. Being able to manage the text size is important in web design. However, you should not use font size adjustments to make paragraphs look like headings, or headings look like paragraphs. Always use the proper HTML tags, like `<h1>` - `<h6>` for headings and `<p>` for paragraphs. The font-size value can be an absolute, or relative size.

Absolute size:

- Sets the text to a specified size
- Does not allow a user to change the text size in all browsers (bad for accessibility reasons)
- Absolute size is useful when the physical size of the output is known

Set Font Size With Pixels

Setting the text size with pixels gives you full control over the text size:

Example

```
h1 {
    font-size: 40px;
}
h2 {
    font-size: 30px;
}
p {
    font-size: 14px;
}
```

Set Font Size With Em

To allow users to resize the text (in the browser menu), many developers use em instead of pixels. The em size unit is recommended by the W3C. 1em is equal to the current font size. The default text size in browsers is 16px. So, the default size of 1em is 16px. The size can be calculated from pixels to em using this formula: $pixels/16=em$

Example

```
h1 {
    font-size: 2.5em; /* 40px/16=2.5em */
}
h2 {
    font-size: 1.875em; /* 30px/16=1.875em */
}
p {
    font-size: 0.875em; /* 14px/16=0.875em */
}
```

In the example above, the text size in em is the same as the previous example in pixels. However, with the em size, it is possible to adjust the text size in all browsers. Unfortunately, there is still a problem with older versions of IE. The text becomes larger than it should when made larger, and smaller than it should when made smaller.

3.18 Use a Combination of Percent and Em

The solution that works in all browsers, is to set a default font-size in percent for the <body> element:

Example

```
body {  
    font-size: 100%;  
}  
h1 {  
    font-size: 2.5em;  
}  
h2 {  
    font-size: 1.875em;  
}  
p {  
    font-size: 0.875em;  
}
```

Font Weight

The `font-weight` property specifies the weight of a font:

```
p.normal {  
    font-weight: normal;  
}  
p.thick {  
    font-weight: bold;  
}
```

Font Variant

The `font-variant` property specifies whether or not a text should be displayed in a small-caps font. In a small-caps font, all lowercase letters are converted to uppercase letters. However, the converted uppercase letters appears in a smaller font size than the original uppercase letters in the text.

Example

```
p.normal {  
    font-variant: normal;  
}  
p.small {  
    font-variant: small-caps;  
}
```

3.19 CSS Links

With CSS, links can be styled in different ways.

Styling Links

Links can be styled with any CSS property (e.g. `color`, `font-family`, `background`, etc.).

Example

```
a {
    color: hotpink;
}
```

In addition, links can be styled differently depending on what **state** they are in.

The four links states are:

- `a:link` - a normal, unvisited link
- `a:visited` - a link the user has visited
- `a:hover` - a link when the user mouses over it
- `a:active` - a link the moment it is clicked

Example

```
/* unvisited link */
a:link {
    color: red;
}
/* visited link */
a:visited {
    color: green;
}
/* mouse over link */
a:hover {
    color: hotpink;
}
/* selected link */
a:active {
    color: blue;
}
```

When setting the style for several link states, there are some order rules:

- `a:hover` MUST come after `a:link` and `a:visited`
- `a:active` MUST come after `a:hover`

Text Decoration

The `text-decoration` property is mostly used to remove underlines from links:

```
a:link {
    text-decoration: none;
}
a:visited {
    text-decoration: none;
}
a:hover {
    text-decoration: underline;
}
a:active {
    text-decoration: underline;
}
```

Background Color

The `background-color` property can be used to specify a background color for links:

```
a:link {
    background-color: yellow;
}
a:visited {
    background-color: cyan;
}
a:hover {
    background-color: lightgreen;
}
a:active {
    background-color: hotpink;
}
```

Advanced - Link Buttons

This example demonstrates a more advanced example where we combine several CSS properties to display links as boxes/buttons:

```
a:link, a:visited {
    background-color: #f44336;
    color: white;
    padding: 14px 25px;
    text-align: center;
    text-decoration: none;
}
```

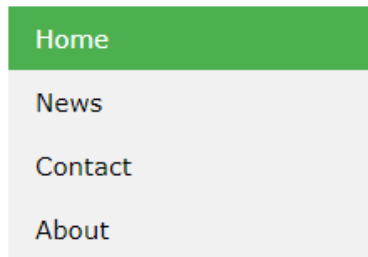
```

    display: inline-block;
}
a:hover, a:active {
    background-color: red;
}

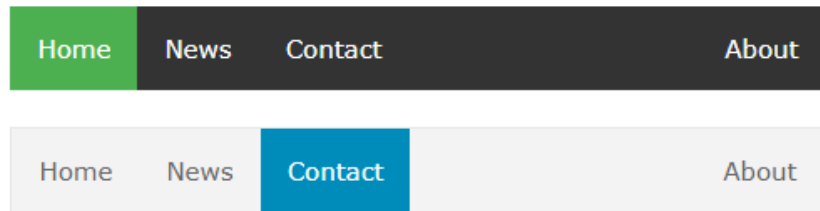
```

CSS Navigation Bar

Vertical



Horizontal



Navigation Bars

With CSS you can transform boring HTML menus into good-looking navigation bars.

Navigation Bar = List of Links

A navigation bar is basically a list of links, so using the `<ul>` and `<li>` elements makes perfect sense:

Example

```

<ul>
  <li><a href="default.asp">Home</a></li>
  <li><a href="news.asp">News</a></li>
  <li><a href="contact.asp">Contact</a></li>
  <li><a href="about.asp">About</a></li>
</ul>

```

- [Home](#)
- [News](#)
- [Contact](#)
- [About](#)

Now let's remove the bullets and the margins and padding from the list:

```
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
}
```

In this example, we remove the bullets from the list, and its default padding and margin.

[Home](#)
[News](#)
[Contact](#)
[About](#)

Example explained:

- `list-style-type: none;` - Removes the bullets. A navigation bar does not need list markers
- Set `margin: 0;` and `padding: 0;` to remove browser default settings

The code in the example above is the standard code used in both vertical and horizontal navigation bars.

Vertical Navigation Bar

To build a vertical navigation bar, you can style the `<a>` elements inside the list, in addition to the code above:

```
li a {
  display: block;
  width: 60px;
}
```

Example explained:

- `display: block;` - Displaying the links as block elements makes the whole link area clickable (not just the text), and it allows us to specify the width (and padding, margin, height, etc. if you want)
- `width: 60px;` - Block elements take up the full width available by default. We want to specify a 60 pixels width

You can also set the width of `<ul>`, and remove the width of `<a>`, as they will take up the full width available when displayed as block elements. This will produce the same result as our previous example:

```

ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    width: 60px;
}
li a {
    display: block;
}
<!DOCTYPE html>
<html>
<head>
<style>
ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    width: 60px;
}

li a {
    display: block;
    background-color: #dddddd;
}
</style>
</head>
<body>

<ul>
    <li><a href="#home">Home</a></li>
    <li><a href="#news">News</a></li>
    <li><a href="#contact">Contact</a></li>
    <li><a href="#about">About</a></li>
</ul>

<p>A background color is added to the links to show the link
area.</p>
<p>Notice that the whole link area is clickable, not just the
text.</p>

</body>
</html>

```



A background color is added to the links to show the link area.

Notice that the whole link area is clickable, not just the text.

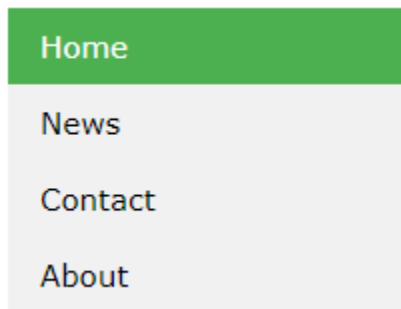
Vertical Navigation Bar Examples

Create a basic vertical navigation bar with a gray background color and change the background color of the links when the user moves the mouse over them:



Active/Current Navigation Link

Add an "active" class to the current link to let the user know which page he/she is on:



```

<!DOCTYPE html>
<html>
<head>
<style>
ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    width: 200px;
    background-color: #f1f1f1;
}

li a {
    display: block;
    color: #000;
    padding: 8px 16px;
    text-decoration: none;
}

li a.active {
    background-color: #4CAF50;
    color: white;
}

li a:hover:not(.active) {
    background-color: #555;
    color: white;
}
</style>
</head>
<body>

<h2>Vertical Navigation Bar</h2>
<p>In this example, we create an "active" class with a green background color and a white text. The class is added to the "Home" link.</p>

<ul>
<li><a class="active" href="#home">Home</a></li>
<li><a href="#news">News</a></li>
<li><a href="#contact">Contact</a></li>
<li><a href="#about">About</a></li>
</ul>

</body>
</html>

```

Vertical Navigation Bar

In this example, we create an "active" class with a green background color and a white text. The class is added to the "Home" link.



Vertical Navigation Bar Examples

Create a basic vertical navigation bar with a gray background color and change the background color of the links when the user moves the mouse over them:



Example

```
ul {  
  list-style-type: none;  
  margin: 0;  
  padding: 0;  
  width: 200px;  
  background-color: #f1f1f1;  
}  
li a {  
  display: block;  
  color: #000;  
  padding: 8px 16px;  
  text-decoration: none;  
}  
/* Change the link color on hover */  
li a:hover {  
  background-color: #555;  
  color: white;  
}
```

```

<!DOCTYPE html>
<html>
<head>
<style>
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  width: 200px;
  background-color: #f1f1f1;
}

li a {
  display: block;
  color: #000;
  padding: 8px 16px;
  text-decoration: none;
}

/* Change the link color on hover */
li a:hover {
  background-color: #555;
  color: white;
}
</style>
</head>
<body>

<h2>Vertical Navigation Bar</h2>

<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>

</body>
</html>

```

Vertical Navigation Bar

Home
News
Contact
About

Center Links & Add Borders

Add `text-align: center` to `<li>` or `<a>` to center the links. Add the `border` property to `<ul>` add a border around the navbar. If you also want borders inside the navbar, add a `border-bottom` to all `<li>` elements, except for the last one:

Home
News
Contact
About

Example

```
ul {
  border: 1px solid #555;
}
li {
  text-align: center;
  border-bottom: 1px solid #555;
}
li:last-child {
  border-bottom: none;
}
```

```
<!DOCTYPE html>
<html><head>
<style>
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  width: 200px;
  background-color: #f1f1f1;
  border: 1px solid #555;
}
li a {
  display: block;
  color: #000;
  padding: 8px 16px;
  text-decoration: none;
}
li {
  text-align: center;
  border-bottom: 1px solid #555;
}
li:last-child {
  border-bottom: none;
}
li a.active {
  background-color: #4CAF50;
  color: white;
}
li a:hover:not(.active) {
  background-color: #555;
  color: white;
}
</style></head><body>
<h2>Vertical Navigation Bar</h2>
<p>we center the navigation links and add a border to the navigation
bar.</p>
<ul>
<li><a class="active" href="#home">Home</a></li>
<li><a href="#news">News</a></li>
<li><a href="#contact">Contact</a></li>
<li><a href="#about">About</a></li>
</ul></body></html>
```

Home
News
Contact
About

Horizontal Navigation Bar

There are two ways to create a horizontal navigation bar. Using **inline** or **floating** list items.

Inline List Items

One way to build a horizontal navigation bar is to specify the `<li>` elements as inline, in addition to the "standard" code above:

Example

```
li {
    display: inline;
}
```

Example explained:

- `display: inline;` - By default, `<li>` elements are block elements. Here, we remove the line breaks before and after each list item, to display them on one line

Floating List Items

Another way of creating a horizontal navigation bar is to float the `<li>` elements, and specify a layout for the navigation links:

Example

```
li {
    float: left;
}
a {
    display: block;
    padding: 8px;
    background-color: #dddddd;
}
```

Example explained:

- `float: left;` - use float to get block elements to slide next to each other
- `display: block;` - Displaying the links as block elements makes the whole link area clickable (not just the text), and it allows us to specify padding (and height, width, margins, etc. if you want)
- `padding: 8px;` - Since block elements take up the full width available, they cannot float next to each other. Therefore, specify some padding to make them look good
- `background-color: #dddddd;` - Add a gray background-color to each a element

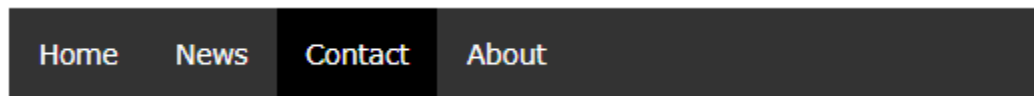
Tip: Add the background-color to instead of each <a> element if you want a full-width background color:

Example

```
ul {
    background-color: #dddddd;
}
```

Horizontal Navigation Bar Examples

Create a basic horizontal navigation bar with a dark background color and change the background color of the links when the user moves the mouse over them:



Example

```
ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    overflow: hidden;
    background-color: #333;
}
li {
    float: left;
}
li a {
    display: block;
    color: white;
    text-align: center;
    padding: 14px 16px;
    text-decoration: none;
}
/* Change the link color to #111 (black) on hover */
li a:hover {
    background-color: #111;
}
```

```

<!DOCTYPE html>
<html>
<head>
<style>
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  overflow: hidden;
  background-color: #333;
}
li {
  float: left;
}
li a {
  display: block;
  color: white;
  text-align: center;
  padding: 14px 16px;
  text-decoration: none;
}
li a:hover {
  background-color: #111;
}
</style>
</head>
<body>

<ul>
  <li><a class="active" href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>

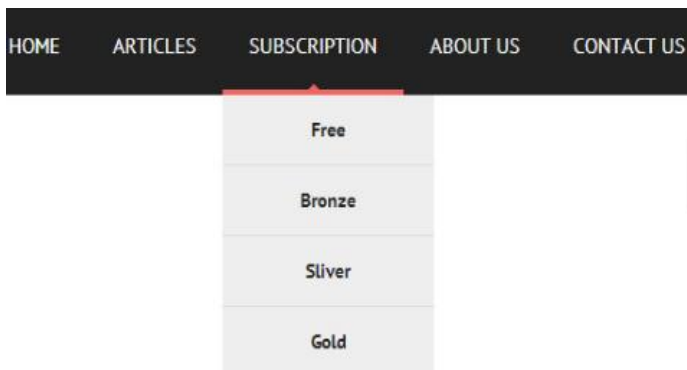
</body>
</html>

```

Home News Contact About

Dropdown Menu

Create a dropdown menu that allows the user to choose an option from a list:



```

<style>
/* Style The Dropdown Button */
.dropbtn {
  background-color: #4CAF50;
  color: white;
  padding: 16px;
  font-size: 16px;
  border: none;
  cursor: pointer;
}
/* The container <div> - needed to position the dropdown content */
.dropdown {
  position: relative;
  display: inline-block;
}
/* Dropdown Content (Hidden by Default) */
.dropdown-content {
  display: none;
  position: absolute;
  background-color: #f9f9f9;
  min-width: 160px;
  box-shadow: 0px 8px 16px 0px rgba(0,0,0,0.2);
  z-index: 1;
}
/* Links inside the dropdown */
.dropdown-content a {
  color: black;
  padding: 12px 16px;
  text-decoration: none;
  display: block;}
/* Change color of dropdown links on hover */
.dropdown-content a:hover {background-color: #f1f1f1}
/* Show the dropdown menu on hover */
.dropdown:hover .dropdown-content {
  display: block;}
/* Change the background color of the dropdown button when the dropdown content is
shown */
.dropdown:hover .dropbtn {
  background-color: #3e8e41; }
</style>
<div class="dropdown">
  <button class="dropbtn">Dropdown</button>
  <div class="dropdown-content">
    <a href="#">Link 1</a>
    <a href="#">Link 2</a>
    <a href="#">Link 3</a>
  </div>
</div>

```

```

</div>
</div>

```

```

<!DOCTYPE html>
<html>
<head>
<style>
.dropbtn {
  background-color: #4CAF50;
  color: white;
  padding: 16px;
  font-size: 16px;
  border: none;
  cursor: pointer;
}

.dropdown {
  position: relative;
  display: inline-block;
}

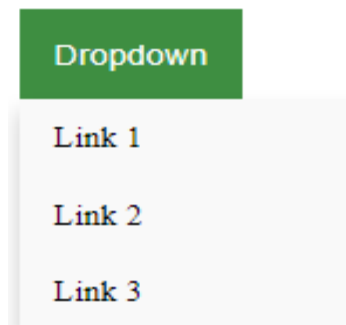
.dropdown-content {
  display: none;
  position: absolute;
  background-color: #f9f9f9;
  min-width: 160px;
  box-shadow: 0px 8px 16px 0px rgba(0,0,0,0.2);
  z-index: 1;
}

.dropdown-content a {
  color: black;
  padding: 12px 16px;
  text-decoration: none;
  display: block;
}

.dropdown-content a:hover {background-color: #f1f1f1}

.dropdown:hover .dropdown-content {
  display: block;
}

```



```

.dropdown:hover .dropbtn {
    background-color: #3e8e41;
}
</style>
</head>
<body>

<h2>Dropdown Menu</h2>
<p>Move the mouse over the button to open the dropdown menu.</p>

<div class="dropdown">
  <button class="dropbtn">Dropdown</button>
  <div class="dropdown-content">
    <a href="#">Link 1</a>
    <a href="#">Link 2</a>
    <a href="#">Link 3</a>
  </div>
</div>

<p><strong>Note:</strong> We use href="#" for test links. In a real web
site this would be URLs.</p>

</body>
</html>

```

Right-aligned Dropdown Content



If you want the dropdown menu to go from right to left, instead of left to right, add `right: 0;`

Example

```

.dropdown-content {
    right: 0;
}

```

```

}
<!DOCTYPE html>
<html>
<head>
<style>
.dropbtn {
  background-color: #4CAF50;
  color: white;
  padding: 16px;
  font-size: 16px;
  border: none;
  cursor: pointer;
}

.dropdown {
  position: relative;
  display: inline-block;
}

.dropdown-content {
  display: none;
  position: absolute;
  right: 0;
  background-color: #f9f9f9;
  min-width: 160px;
  box-shadow: 0px 8px 16px 0px rgba(0,0,0,0.2);
  z-index: 1;
}

.dropdown-content a {
  color: black;
  padding: 12px 16px;
  text-decoration: none;
  display: block;
}

.dropdown-content a:hover {background-color: #f1f1f1}

.dropdown:hover .dropdown-content {
  display: block;
}

```

```

.dropdown:hover .dropbtn {
  background-color: #3e8e41;
}
</style>
</head>
<body>

<h2>Aligned Dropdown Content</h2>
<p>Determine whether the dropdown content should go from left to right
or right to left with the left and right properties.</p>

<div class="dropdown" style="float:left;">
  <button class="dropbtn">Left</button>
  <div class="dropdown-content" style="left:0;">
    <a href="#">Link 1</a>
    <a href="#">Link 2</a>
    <a href="#">Link 3</a>
  </div>
</div>

<div class="dropdown" style="float:right;">
  <button class="dropbtn">Right</button>
  <div class="dropdown-content">
    <a href="#">Link 1</a>
    <a href="#">Link 2</a>
    <a href="#">Link 3</a>
  </div>
</div>

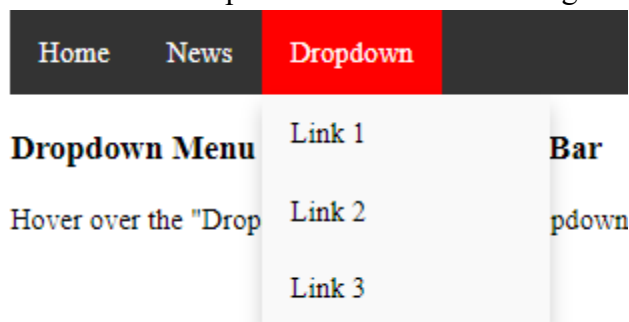
</body>
</html>

```



Dropdown Navbar

How to add a dropdown menu inside a navigation bar.



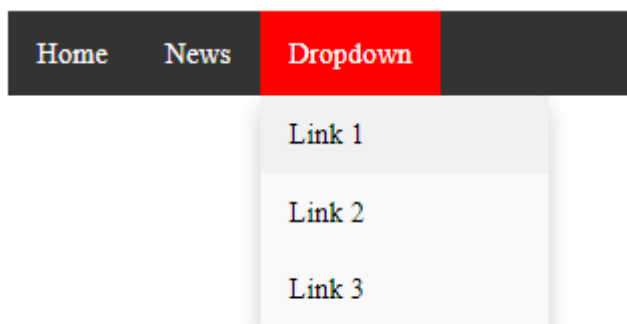
Dropdown Menu inside a Navigation Bar

Hover over the "Dropdown" link to see the dropdown menu.

```

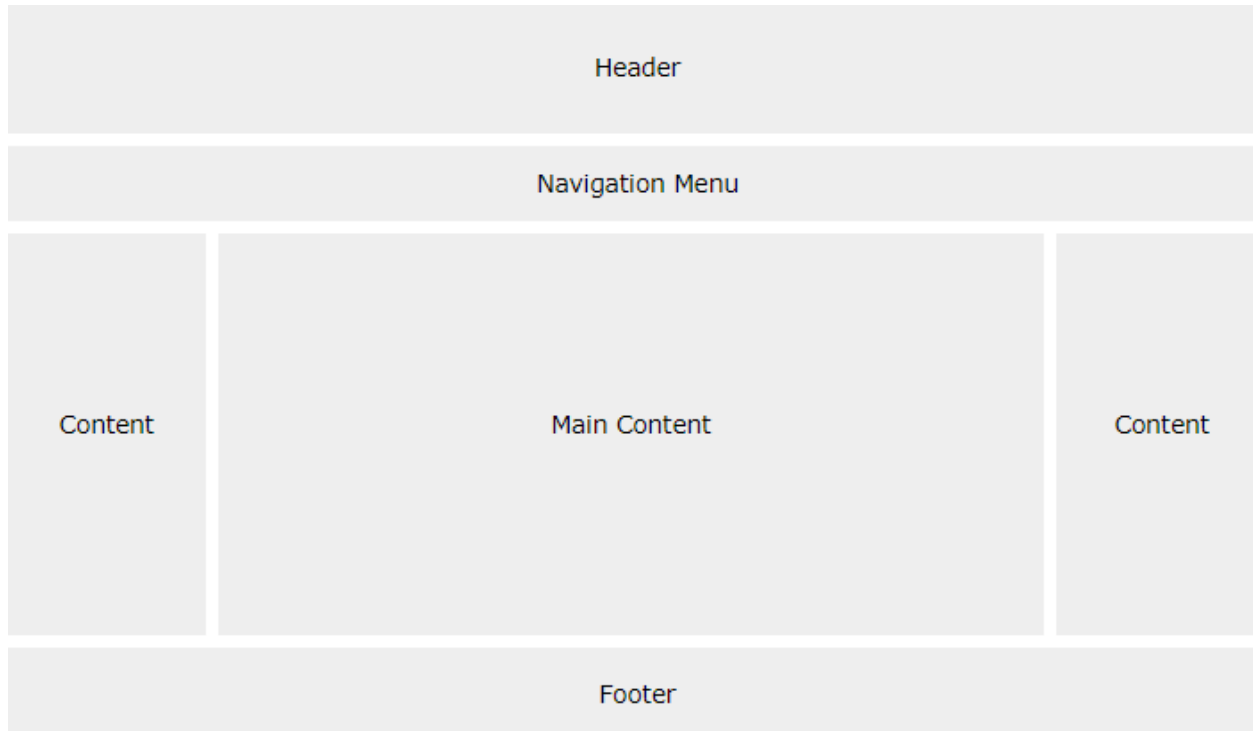
<!DOCTYPE html>
<html>
<head>
<style>
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  overflow: hidden;
  background-color: #333;
}
li {
  float: left;
}
li a, .dropbtn {
  display: inline-block;
  color: white;
  text-align: center;
  padding: 14px 16px;
  text-decoration: none;
}
li a:hover, .dropdown:hover .dropbtn {
  background-color: red;
}
li.dropdown {
  display: inline-block;
}
.dropdown-content {
  display: none;
  position: absolute;
  background-color: #f9f9f9;
  min-width: 160px;
  box-shadow: 0px 8px 16px 0px rgba(0,0,0,0.2);
  z-index: 1;
}
.dropdown-content a {
  color: black;
  padding: 12px 16px;
  text-decoration: none;
  display: block;
  text-align: left;
}
.dropdown-content a:hover {background-color: #f1f1f1}
.dropdown:hover .dropdown-content {
  display: block;
}
</style>
</head>
<body>
<ul>
<li><a href="#home">Home</a></li>
<li><a href="#news">News</a></li>
<li class="dropdown">
  <a href="javascript:void(0)" class="dropbtn">Dropdown</a>
  <div class="dropdown-content">
    <a href="#">Link 1</a>
    <a href="#">Link 2</a>
    <a href="#">Link 3</a>
  </div>
</li>
</ul>
</body>
</html>

```



Website Layout

A website is often divided into headers, menus, content, and footer:



Header

A header is usually located at the top of the website (or right below a top navigation menu). It often contains a logo or the website name:

Example

```
.header {  
  background-color: #F1F1F1;  
  text-align: center;  
  padding: 20px;  
}
```

Navigation Bar

A navigation bar contains a list of links to help visitors navigating through your website:

Example

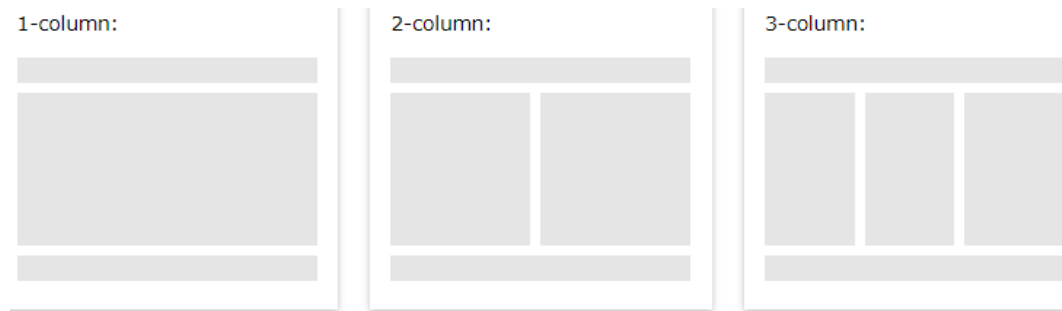
```
/* The navbar container */
.menu {
    overflow: hidden;
    background-color: #333;
}
/* Navbar links */
.menu a {
    float: left;
    display: block;
    color: #f2f2f2;
    text-align: center;
    padding: 14px 16px;
    text-decoration: none;
}
/* Links - change color on hover */
.topnav a:hover {
    background-color: #ddd;
    color: black;
}
```



Content

The layout in this section, often depends on the target users. The most common layout is one (or combining them) of the following:

- **1-column** (often used for mobile browsers)
- **2-column** (often used for tablets and laptops)
- **3-column layout** (only used for desktops)



Example

```
/* Create three equal columns that floats next to each other */
.column {
    float: left;
    width: 33.33%;
}
/* Clear floats after the columns */
.row:after {
    content: "";
    display: table;
    clear: both;
}
/* Responsive layout - makes the three columns stack on top of each other
instead of next to each other on smaller screens (600px wide or less) */
@media (max-width: 600px) {
    .column {
        width: 100%;
    }
}
```

Footer

The footer is placed at the bottom of your page. It often contains information like copyright and contact info:

Example

```
.footer {
    background-color: #F1F1F1;
    text-align: center;
    padding: 10px;
}
```

Responsive Web Design (RWD) – Introduction (Take-home Assignment)

Reference - https://www.w3schools.com/css/css_rwd_intro.asp

1. *RWD Viewport*
2. *RWD Grid View*
3. *RWD Media Queries*
4. *RWD Images*
5. *RWD Videos*

What is Responsive Web Design?

- Responsive web design makes your web page look good on all devices.
- Responsive web design uses only HTML and CSS.
- Responsive web design is not a program or a JavaScript.

Designing For the Best Experience for All Users

Web pages can be viewed using many different devices: desktops, tablets, and phones. Your web page should look good, and be easy to use, regardless of the device.

Web pages should not leave out information to fit smaller devices, but rather adapt their content to fit any device:



It is called responsive web design when you use CSS and HTML to resize, hide, shrink, enlarge, or move the content to make it look good on any screen.